9.4 Sequences as Functions

WORKSHEET

The question

STEP I 2016, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s1-q8

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself What sequence has generating function $\frac{1}{1-x}$? What about $\frac{1}{1-2x}$? What happens to the sequence if you multiply its generating function by x?

STAGE 1 The natural numbers and the triangle numbers

Covers: Part (i)

Ask yourself Expand $(1-x)^{-2}$ and $(1-x)^{-3}$ by the binomial series. What is the coefficient of x^n in each? What does multiplying by x do?

Pause and try Write n^2 using n and $\frac{1}{2}n(n+1)$. Generating functions add just as sequences do.

STAGE 2 A recurrence becomes an equation

Covers: Part (ii)(a)

Pause and try Multiply $u_n = ku_{n-1}$ by x^n and add over all $n \geq 1$. Write each side in terms of $f(x)$, remembering that the left-hand sum is missing the $n = 0$ term.

STAGE 3 The Fibonacci numbers

Covers: Part (ii)(b)

Ask yourself Do the same with $u_n = u_{n-1} + u_{n-2}$, summing over $n \geq 2$. Which early terms are missing from each sum? Then use $u_0 = 0$ and $u_1 = 1$.

STAGE 4 Explaining $1/89$

Covers: beyond the question

Ask yourself Put $x = \frac{1}{10}$ in your answer to Part (ii)(b). What do you notice?

What's really going on

From generating function to formula

Generating functions do not just encode a sequence: they can solve for it. The denominator factorises as $1 - x - x^2 = (1 - \varphi x)(1 - \psi x)$, where $\varphi = \frac{1+\sqrt{5}}{2}$ and $\psi = \frac{1-\sqrt{5}}{2}$ are the roots of $t^2 = t + 1$. Partial fractions give

$$\frac{x}{1 - x - x^2} = \frac{1}{\sqrt{5}}\left(\frac{1}{1 - \varphi x} - \frac{1}{1 - \psi x}\right),$$

and expanding each fraction as a geometric series,

$$F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}.$$

This is *Binet's formula*. Since $|\psi| < 1$, the second term is tiny, so F_n is the nearest integer to $\varphi^n/\sqrt{5}$, and the ratio of successive Fibonacci numbers tends to the golden ratio (the

subject of Section 2.2). The same method solves any linear recurrence with constant coefficients; the exercises include an example from STEP I 2004.

Why generating functions are so useful

The real power appears when we multiply. The coefficient of x^n in a product $f(x)g(x)$ is $\sum_k u_k v_{n-k}$, which adds up over all ways of splitting n into two parts. That is exactly what we need to count things built from two pieces. The next two sections use it to count the ways of making change (Section 9.5) and to find strangely numbered dice with the same totals as ordinary ones (Section 9.6). In probability, the same idea gives the *probability generating function*, which turns sums of independent random variables into products.

De Moivre, Euler and Laplace

Abraham de Moivre used generating functions in the 1720s to solve linear recurrences, essentially by the method above. Leonhard Euler used them in 1748 to study partitions of whole numbers, and Pierre-Simon Laplace made them the foundation of his *Théorie analytique des probabilités* (1812). In the twentieth century they became central to combinatorics, and analytic methods (studying a generating function as a function of a complex variable) now give precise estimates for counting problems far too complicated to solve exactly.

Try this next

1. **Two digits at a time.** Try STEP II 2022, Question 3 (in OCR's download of the 2022 papers, www.ocr.org.uk/Images/696587-2022-past-papers.zip). It explains why the decimal expansion of $\frac{1}{89}$ begins with the Fibonacci numbers, and asks for a fraction whose decimals show the pattern two digits at a time.

2. **Solving a recurrence.** Try STEP I 2004, Question 7 (step.maths.org/questions/04-s1-q7). It solves a linear recurrence with a generating function and partial fractions, then runs the method in reverse.

3. **Counting numbers.** Show that $\frac{1}{9801} = 0.00\,01\,02\,03\,04\ldots$, with the two-digit blocks counting upwards. Where does the pattern first go wrong, and why?

4. **Sums of squares.** Explain why $\frac{1}{1-x}f(x)$ is the generating function of the partial sums $u_0 + u_1 + \cdots + u_n$. Use this and the answer to part (i) to find a formula for $1^2 + 2^2 + \cdots + n^2$.

Further reading

- **Herbert Wilf, *generatingfunctionology*** (A K Peters; free online). The classic introduction. Chapter 1 is perfect after this section.

- **Ronald Graham, Donald Knuth & Oren Patashnik, *Concrete Mathematics*** (Addison-Wesley), Chapter 7.
- **Philippe Flajolet & Robert Sedgewick, *Analytic Combinatorics*** (Cambridge University Press; free online). Far beyond first year, but the opening pages show where the subject leads.

The full guided solution is in Section 9.4:
https://beyondstep.keithmcnulty.org/ch09/generating-functions.html